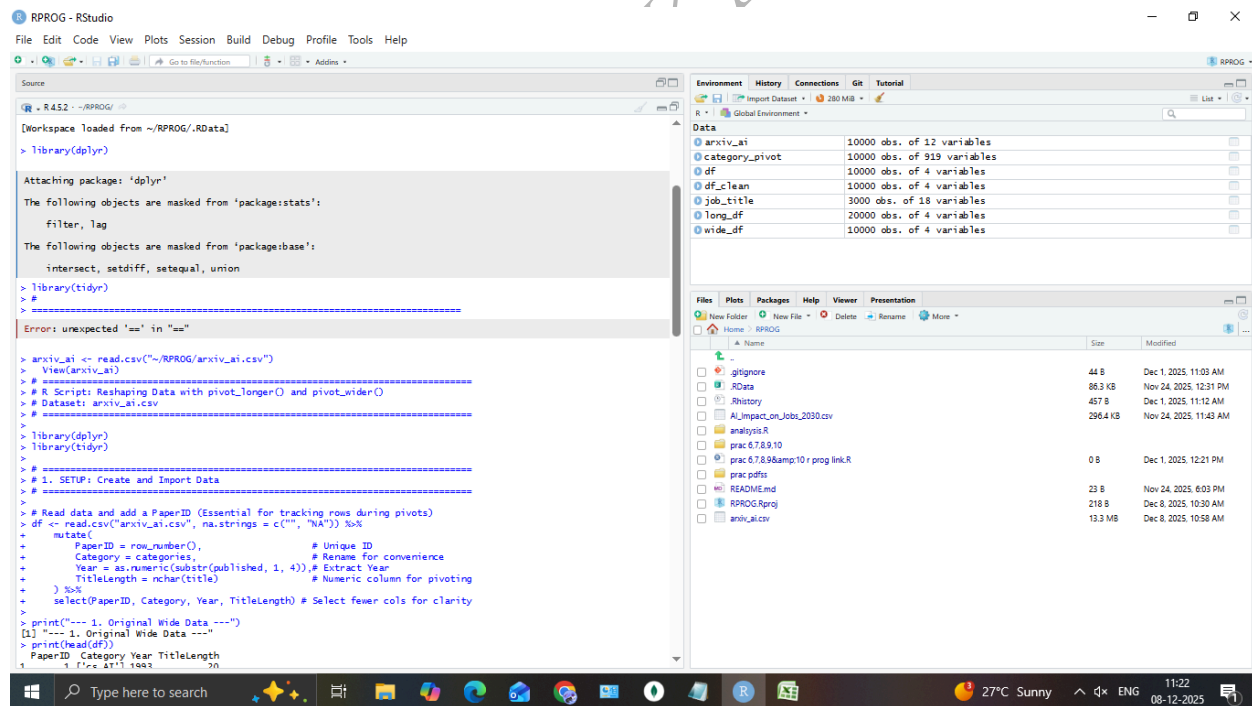


SHETH L.U.J AND SIR M.V. COLLEGE

SUBJECT NAME: Data Analysis with SAS / SPSS / R

PRACTICAL 11

AIM: Reshaping data using
pivot_longer()/pivot_wider() (R).



The screenshot shows the RStudio interface with the following components:

- Source Pane:** Contains R code for loading the 'dplyr' package, reading a CSV file, and performing data reshaping operations using 'pivot_longer()' and 'pivot_wider()'.
- Environment Pane:** Lists the objects in the R environment, including 'arxiv_ai', 'category_pivot', 'df', 'df_clean', 'job_title', 'long_df', and 'wide_df'.
- Files Pane:** Shows the file explorer with various files and folders, including 'RData', 'Rhistory', 'AI_Impact_on_Job_2030.csv', 'analysis.R', 'prac 6.7.8.9.10', 'prac pdfs', 'README.md', 'RPROG.Rproj', and 'arxiv_ai.csv'.

```
RStudio - RPROG - RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Workspace loaded from ~/RPROG/RData
> library(dplyr)

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':
  filter, lag

The following objects are masked from 'package:base':
  intersect, setdiff, setequal, union

> library(tidyverse)
> #
Error: unexpected '=' in '='
> arxiv_ai <- read.csv("~/RPROG/arxiv_ai.csv")
> View(arxiv_ai)
> #
> # R Script: Reshaping Data with pivot_longer() and pivot_wider()
> # Dataset: arxiv_ai.csv
> #
> library(dplyr)
> library(tidyverse)
> #
> # =====
> # 1. SETUP: Create and Import Data
> # =====
> # Read data and add a PaperID (Essential for tracking rows during pivots)
> df <- read.csv("arxiv_ai.csv", na.strings = c("", "NA")) %>%
+   mutate(
+     PaperID = row_number(), # Unique ID
+     Category = categories, # Rename for convenience
+     Year = as.numeric(substr(published, 1, 4)), # Extract Year
+     TitleLength = nchar(title) # Numeric column for pivoting
+   ) %>%
+   select(PaperID, Category, Year, TitleLength) # Select fewer cols for clarity
>
> print("---- 1. Original Wide Data ----")
[1] "---- 1. Original Wide Data ----"
> print(head(df))
  PaperID Category Year TitleLength
1 1796 2711 1993          20
```

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SUBJECT NAME: Data Analysis with SAS / SPSS / R



```
RPROG - RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to File/Function Addins
Source Environment History Connections Git Tutorial
R4.5.2 - RPROG.R
select(PaperID, Category, Year, TitleLength) # Select fewer cols for clarity
>
print("---- 1. Original Wide Data ----")
[1] "---- 1. Original Wide Data ----"
> print(head(df))
  PaperID Category Year TitleLength
1      1 ['cs.AI'] 1993         20
2      2 ['cs.AI'] 1993        105
3      3 ['cs.AI'] 1993         39
4      4 ['cs.AI'] 1993         52
5      5 ['cs.AI'] 1993         69
6      6 ['cs.AI'] 1993         70
>
# =====
# 2. PIVOT_LONGER (Wide to Long)
# =====
>
# Scenario: Combine 'Year' and 'TitleLength' into a single column called 'Value'
# Useful for: Plotting multiple metrics like Year and TitleLength together.
> long_df <- df %>%
+   pivot_longer(
+     cols = c(Year, TitleLength), # Columns we want to stack
+     names_to = "Metric",         # Label for metric names
+     values_to = "Value"         # Values of numeric columns
+   )
>
print("---- 2. Long Format (pivot_longer) ----")
[1] "---- 2. Long Format (pivot_longer) ----"
> # Notice how PaperID 1 now appears TWICE (once for Year, once for TitleLength)
> print(head(long_df, 6))
# A tibble: 6 x 4
  PaperID Category Metric      Value
<int> <chr> <chr> <dbl>
1      1 ['cs.AI'] Year      1993
2      1 ['cs.AI'] TitleLength 20
3      2 ['cs.AI'] Year      1993
4      2 ['cs.AI'] TitleLength 105
5      3 ['cs.AI'] Year      1993
6      3 ['cs.AI'] TitleLength 39
>
# =====
# 3. PIVOT_WIDER (Long to Wide)
# =====
>
# Scenario: Spread the 'Metric' values back into separate columns again.
> wide_df <- long_df %>%
+   pivot_wider(
+     names_from = Metric, # New column headers
+     values_from = Value, # Values from long format
+   )
>
print(head(wide_df, 6))
# A tibble: 6 x 5
  PaperID Category Year TitleLength
<int> <chr> <dbl> <dbl>
1      1 ['cs.AI'] 1993         20
2      2 ['cs.AI'] 1993        105
3      3 ['cs.AI'] 1993         39
4      4 ['cs.AI'] 1993         52
5      5 ['cs.AI'] 1993         69
6      6 ['cs.AI'] 1993         70
>
```

Environment: arkiv_ai (10000 obs. of 12 variables), category_pivot (10000 obs. of 919 variables), df (10000 obs. of 4 variables), df_clean (10000 obs. of 4 variables), job_title (3000 obs. of 18 variables), long_df (20000 obs. of 4 variables), wide_df (10000 obs. of 4 variables)

Files: .gitignore (44 B, Dec 1, 2025, 11:03 AM), RData (86.3 KB, Nov 24, 2025, 12:31 PM), Rhistory (457 B, Dec 1, 2025, 11:12 AM), ALImpact_on_Jobs_2030.csv (296.4 KB, Nov 24, 2025, 11:43 AM), analysis.R, prac.6.7.8.9.10, prac.6.7.8.9&10.r prog link.R, prac.pdfs, README.md (23 B, Nov 24, 2025, 6:03 PM), RPROG.Rproj (218 B, Dec 8, 2025, 10:30 AM), arxiv_ai.csv (13.3 MB, Dec 8, 2025, 10:58 AM)

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SUBJECT NAME: Data Analysis with SAS / SPSS / R



```
RPROG - RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Source
R 4.5.2 - ~/RPROG/
> # Scenario: Spread the 'Metric' values back into separate columns again.
> wide_df <- long_df %>%
+   pivot_wider(
+     names_from = Metric, # New column headers
+     values_from = Value # Values to fill cells
+   )
>
> print("---- 3. Wide Format (Back to Original) ----")
[1] "---- 3. Wide Format (Back to Original) ----"
> print(head(wide_df))
# A tibble: 6 x 4
  PaperID Category Year TitleLength
  <int> <chr> <dbl> <dbl>
1 1 ['cs.AI'] 1993 10
2 2 ['cs.AI'] 1993 105
3 3 ['cs.AI'] 1993 39
4 4 ['cs.AI'] 1993 52
5 5 ['cs.AI'] 1993 69
6 6 ['cs.AI'] 1993 70
>
> # =====
> # 4. ADVANCED EXAMPLE (Reshaping for Reporting)
> # =====
> # Create a table where columns are Categories and rows are PaperIDs
> # showing the 'Year' in the grid.
>
> df_clean <- df %>%
+   mutate(Category = ifelse(is.na(Category), "Unknown", Category))
>
> category_pivot <- df_clean %>%
+   select(PaperID, Category, Year) %>%
+   pivot_wider(
+     names_from = Category, # Column headers become categories
+     values_from = Year # Fill with publication year
+   )
>
> print("---- 4. Category Pivot (Spreading Categories) ----")
[1] "---- 4. Category Pivot (Spreading Categories) ----"
> print(head(category_pivot))
# A tibble: 6 x 919
  PaperID ['cs.AI'] ['cs.AI', 'I.2.8'] ['cs.AI', 'I.1.2; I.2.2'] ['cs.AI', 'I.2.0'] ['cs.AI', 'I.2.8; I.1.2']
  <int> <dbl> <dbl> <dbl> <dbl>
1 1 1993 NA NA NA NA
2 2 1993 NA NA NA NA
3 3 1993 NA NA NA NA
4 4 1993 NA NA NA NA
5 5 1993 NA NA NA NA
6 6 1993 NA NA NA NA
>
Environment History Connections Git Tutorial
Data
  arkiv_ai 10000 obs. of 12 variables
  category_pivot 10000 obs. of 919 variables
  df 10000 obs. of 4 variables
  df_clean 10000 obs. of 4 variables
  job_title 3000 obs. of 18 variables
  long_df 20000 obs. of 4 variables
  wide_df 10000 obs. of 4 variables
Files Plots Packages Help Viewer Presentation
  New Folder New File Delete Rename More
  Home RPROG
  Name Size Modified
  . 44 B Dec 1, 2025, 11:03 AM
  .gitignore 863 KB Nov 24, 2025, 12:31 PM
  .RData 457 B Dec 1, 2025, 11:12 AM
  .Rhistory 296.4 KB Nov 24, 2025, 11:43 AM
  ALImpact_on_Jobs_2030.csv
  analysis.R
  prac.6.7.8.9.10
  prac.6.7.8.9&10.r prog link.R 0 B Dec 1, 2025, 12:21 PM
  prac.pdfs
  README.md 23 B Nov 24, 2025, 6:03 PM
  RPROG.Rproj 218 B Dec 8, 2025, 10:30 AM
  arkiv_ai.csv 13.3 MB Dec 8, 2025, 10:58 AM
Type here to search 27°C Sunny 11:24 08-12-2025
```

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SUBJECT NAME: Data Analysis with SAS / SPSS / R



RPROG - RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Source

```
> print("---- 3. Wide Format (Back to Original) ----")
[1] "---- 3. Wide Format (Back to Original) ----"
> print(head(df))
# A tibble: 6 x 5
  PaperID Year TitleLength
  <dbl>   <dbl>         <dbl>
1 1 ['cs.AI'] 1993         20
2 2 ['cs.AI'] 1993         105
3 3 ['cs.AI'] 1993         39
4 4 ['cs.AI'] 1993         52
5 5 ['cs.AI'] 1993         69
6 6 ['cs.AI'] 1993         70

> # =====
> # 4. ADVANCED EXAMPLE (Reshaping for Reporting)
> # =====

> # Create a table where columns are Categories and rows are PaperIDs
> # showing the 'Year' in the grid.

df_clean <- df %>%
+ mutate(Category = ifelse(is.na(Category), "Unknown", Category))
>
category_pivot <- df_clean %>%
+ select(PaperID, Category, Year) %>%
+ pivot_wider(
+   names_from = Category, # Column headers become categories
+   values_from = Year,    # Fill with publication year
+ )
>
> print("---- 4. Category Pivot (Spreading Categories) ----")
[1] "---- 4. Category Pivot (Spreading Categories) ----"
> print(head(category_pivot))
# A tibble: 6 x 919
  PaperID ['cs.AI'] ['cs.AI', 'I.2.8'] ['cs.AI', 'I.1.2; I.2.2'] ['cs.AI', 'I.2.0'] ['cs.AI', 'I.2.8; I.1.2']
  <dbl>      <dbl>      <dbl>      <dbl>      <dbl>
1 1 1993      NA      NA      NA      NA
2 2 1993      NA      NA      NA      NA
3 3 1993      NA      NA      NA      NA
4 4 1993      NA      NA      NA      NA
5 5 1993      NA      NA      NA      NA
6 6 1993      NA      NA      NA      NA

# i abbreviated names: ['cs.AI', 'I.2.8; I.2.1; J.6; I.2.4; F.2.2']
# i 913 more variables: ['cs.AI', 'I.2'] <dbl>, ['cs.AI', 'D.1.6'] <dbl>, ['cs.AI', 'I.2.3; I.2.7'] <dbl>,
# ['cs.AI', 'I.2.3; I.2.4'] <dbl>, ['cs.AI', 'A.m'] <dbl>, ['cs.AI', 'I.2.3'] <dbl>,
# ['cs.AI', 'F.4.1; I.2.3'] <dbl>, ['cs.AI', 'I.2.4; F.4.1'] <dbl>, ['cs.AI', 'F.4.1; I.2.3; I.2.4'] <dbl>,
# ['cs.AI', 'Artificial intelligence and nonmonotonic reasoning and belief\n revision'] <dbl>,
# ['cs.AI', 'F.4.1; I.2.4; I.2.3'] <dbl>, ['cs.AI', 'I.2.3; I.2.8'] <dbl>, ['cs.AI', 'I.2.4; F.4.1'] <dbl>,
# ['cs.AI', 'F.4.1; I.2.3'] <dbl>, ['cs.AI', 'I.2.4'] <dbl>, ['cs.AI', 'I.2.8; I.2.3; F.4.1'] <dbl>, ...
# i Use `print(names_of_columns())` to see all column names.
```

Environment History Connections Git Tutorial

Data

arxiv_ai	10000 obs. of 12 variables	
category_pivot	10000 obs. of 919 variables	
df	10000 obs. of 4 variables	
df_clean	10000 obs. of 4 variables	
job_title	3000 obs. of 18 variables	
long_df	20000 obs. of 4 variables	
wide_df	10000 obs. of 4 variables	

Files Plots Packages Help Viewer Presentation

Home RPROG

Name	Size	Modified
ignore	44 B	Dec 1, 2025, 11:03 AM
RData	86.3 KB	Nov 24, 2025, 12:31 PM
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prac.6.7.8.9&10.r prog link.R	0 B	Dec 1, 2025, 12:21 PM
prac.pdfits		
README.md	23 B	Nov 24, 2025, 6:03 PM
RPROG.Rproj	218 B	Dec 8, 2025, 10:30 AM
arxiv_ai.csv	13.3 MB	Dec 8, 2025, 10:58 AM

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